

2.1b

Perception

Perceptual Organization & Interpretation

What is Perceptual Organization?

People organize visual sensations into a **gestalt** :: a whole

Our conscious experience is an **integrated whole**

In perception, the **whole may exceed the sum of its parts**

Brains don't just register info, it filters info & **constructs perception**

Form Perception

Figure & Ground

We perceive **figure-ground**, separating faces from **backgrounds**

Perceiving **an object** :: the **figure**, as distinct from its **surroundings** :: the **ground**.

While reading, words are figure, whitespace is ground

Grouping

We organize a figure into a meaningful form

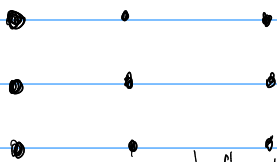
We process basic features, like color, movement & light-dark contrast, instantly & automatically.

To bring order to other stimuli, our mind follows certain rules for grouping.

Proximity - we group nearby figures together

Similarity - we group objects according to how similar they are

Closure - we fill in gaps to create a complete, whole object.



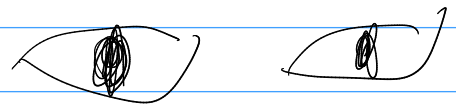
What is **Depth Perception**?

Depth perception :: from 2D images on our retinas we form 3D perceptions

Using a visual cliff, researchers showed that infants could perceive depth.

Crawling increases infant's awareness of heights.

Binocular Cues



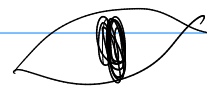
w/ binocular cues we perceive depth

w/ BC we judge distance of nearby objects

Convergence is a BC where retinal images are combined

Retinal Disparity is a BC using the slight differences between retinal images to judge closeness.
Greater disparity, Closer the Object.

Monocular Cues



At big distances, slight binocular differences make little impact.

So we use monocular cues.

Relative Clarity: More light passes thru far objects so we perceive them as hazy

Relative Size: Two Objects similar size, most perceive smaller retinal image as further

Texture gradient: less detailed texture, further

Linear Perspective: Parallel lines appear to meet in distance.
Sharper θ convergence $\rightarrow$ greater distance

Interposition: Object blocks view of another, we see it as closer

Motion Perception

Brain computes motion by assuming:
shrinking objects are retreating
enlarging objects are approaching

large & small objects move at same speed

$\Rightarrow$ larger object seems slower

Apparent Movement:

As we move, stable objects also seem to move.

Fix gaze on some point -

- objects beyond fixation point move w/u
- objects in front of fixation point move backward

Stroboscopic Movement:

brain perceives rapid series of slightly varying images as continuous movement.

Phi Phenomenon:

We perceive two adjacent stationary lights blinking as 1 jumping back and forth.

Auto kinetic effect:

we stare at stationary light in dark room, natural eye movement makes the light seem to move

What is Perceptual Constancy?

Perceptual Constancy: regardless of viewing angle, distance & illumination, we can near-instantly identify people & things.

Constancies:

Color Constancy. We see Color because of brain's computations of light reflected by an object relative to it.

Brightness constancy also depends on context. This perception is dependant on relative luminance $\therefore$ amount of light an object reflects relative to surroundings.

Shape Constancy ensures we perceive the form of familiar objects even as their images change. Because visual cortex neurons rapidly learn to associate different views of an object.

Size constancy allows us to perceive unchanging size, even as distance varies.

What's **Experience & Visual perception**?

There is a critical period where exposure to certain stimuli or experiences is required. Without this, there may be irreparable damage.

What is Perceptual Adaptation?

We can quickly adapt to lenses that significantly distort vision, but there's an aftereffect after removing them, but you'd quickly adapt again.

This is because of our Perceptual Adaptation to the changed sensory input.

MCQs

1b 2. a 3 d. 4a

5b 6. b 7c 8 d

Personal Connection

Throughout the day, I change the color of my smart light bulbs often.

Because of color constancy, I see my things as the same